SYLLABUS

October 09, 2026

Course ID, Course Name, Term
DPAS 625 Clinical Pharmacology I, Fall 2026

Author(s) of syllabus
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Course Description
This is the first course of sequence. This course covers general pharmacologic principles, drug receptor sites, physiologic reactions, half-life, therapeutic effects, metabolism, excretion and possible side effects of different classes of drugs on various organ systems. Practical clinical application is emphasized; drug classes are synchronized with organ systems in General Medicine I.

Course Goal (required by ARC-PA, B1.03d)
By the end of this course, students will integrate foundational pharmacologic principles with therapeutic reasoning to select, evaluate, and communicate safe, effective, evidence-based pharmacotherapy for common conditions encountered in clinical practice.

Credit Hours
3 credit hours

Pre-requisites and Co-requisites
Pre-requisites: All Summer 1 courses
Co-requisites: All Fall 1 courses

Number of contact hours per week
Tuesday and Thursday from for 2.5 hours

Course Learning Outcomes

1. Explain foundational principles of pharmacology and their relationship to medication effects. (KNOW4; B2.02d)
2. Compare pharmacologic agents based on their therapeutic and pharmacologic characteristics. (KNOW4, REASON1; B2.02d, B2.02e)
3. Apply pharmacologic knowledge to select safe and effective pharmacotherapy for common medical conditions. (KNOW4, REASON1; B2.02d)
4. Apply principles of safe medication prescribing and management. (KNOW4; B2.02d)
5. Communicate medication-related information effectively to patients and caregivers. (CLIN4, COMMO1, COMMO2; B2.02d, B2.04, B2.12a)

Instructional Objectives

Instructional Objectives with Learning Outcomes:

Instructional Objectives *On successful completion of this course students will…*	***Associated Learning Outcomes***
Introduction to Pharmacology	
IO1.1 Define and describe fundamental pharmacologic concepts, including pharmacokinetics, pharmacodynamics, drug-receptor interactions, dose-response relationships, therapeutic index, half-life, and steady state.	CLO1
IO1.2 Describe the processes of drug absorption, distribution, metabolism, and excretion and factors that influence these processes.	CLO1
IO1.3 Explain how patient-specific factors, including age, pregnancy, renal and hepatic function, and pharmacogenetic variation, may affect medication response.	CLO1, CLO2
IO1.4 Identify common types and mechanisms of adverse drug reactions, drug-drug interactions, and drug-disease interactions.	CLO2
IO1.5 Interpret common components of medication orders and prescriptions, including drug name, dose, route, frequency, and duration.	CLO4
Musculoskeletal Pharmacotherapeutics	
IO2.1 Identify major pharmacologic classes used in the management of musculoskeletal conditions and describe their mechanisms of action.	CLO2
IO2.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat musculoskeletal conditions.	CLO2
IO2.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.	CLO2
IO2.4 Select appropriate pharmacologic therapy for common musculoskeletal conditions based on clinical presentation and patient-	CLO3

specific factors.	
IO2.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in musculoskeletal conditions.	CLO4
IO2.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for musculoskeletal conditions.	CLO5
Gastrointestinal & Nutrition Pharmacotherapeutics	
IO3.1 Identify major pharmacologic classes used in the management of gastrointestinal and nutrition-related conditions and describe their mechanisms of action.	CLO2
IO3.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat gastrointestinal and nutrition-related conditions.	CLO2
IO3.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.	CLO2
IO3.4 Select appropriate pharmacologic therapy for common gastrointestinal and nutrition-related conditions based on clinical presentation and patient-specific factors.	CLO3
IO3.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in gastrointestinal and nutrition-related conditions.	CLO4
IO3.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for gastrointestinal and nutrition-related conditions.	CLO5
Endocrine Pharmacotherapeutics	
IO4.1 Identify major pharmacologic classes used in the management of endocrine conditions and describe their mechanisms of action.	CLO2
IO4.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat endocrine conditions.	CLO2
IO4.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.	CLO2
IO4.4 Select appropriate pharmacologic therapy for common endocrine conditions based on clinical presentation and patient-specific factors.	CLO3
IO4.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in endocrine conditions.	CLO4
IO4.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for endocrine conditions.	CLO5
Renal Pharmacotherapeutics	
IO5.1 Identify major pharmacologic classes used in the management of renal conditions and describe their mechanisms of action.	CLO2
IO5.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat renal conditions.	CLO2

IO5.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.	CLO2
IO5.4 Select appropriate pharmacologic therapy for common renal conditions based on clinical presentation and patient-specific factors.	CLO3
IO5.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in renal conditions.	CLO4
IO5.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for renal conditions.	CLO5
Genitourinary & Reproductive Pharmacotherapeutics	
IO6.1 Identify major pharmacologic classes used in the management of genitourinary and reproductive conditions s and describe their mechanisms of action.	CLO2
IO6.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat genitourinary conditions and reproductive medicine.	CLO2
IO6.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.	CLO2
IO6.4 Select appropriate pharmacologic therapy for common genitourinary & reproductive conditions based on clinical presentation and patient-specific factors.	CLO3
IO6.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in genitourinary & reproductive conditions.	CLO4
IO6.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for genitourinary & reproductive conditions.	CLO5

Method of Assessing Student Achievement

Table of Learning Outcomes with Assessment/Learning Activities:

Course Objectives/Student Learning Outcomes *On successful completion of this course, students will…*	***Associated Learning Assessment Activities for each learning outcome***
1. Explain foundational principles of pharmacology and their relationship to medication effects.	Unit Examinations, In-class quizzes, Discussion Board assignments, Clinical Case assignments and Comprehensive Final
2. Compare pharmacologic agents based on their therapeutic and pharmacologic characteristics.	Unit Examinations, In-class quizzes, Discussion Board, assignments, Clinical Case assignments and Comprehensive Final
3. Apply pharmacologic knowledge to select safe and effective pharmacotherapy for common medical conditions.	Unit Examinations, In-class quizzes, Discussion Board assignments, Clinical Case assignments and Comprehensive Final

Course Objectives/Student Learning Outcomes *On successful completion of this course, students will…*	***Associated Learning Assessment Activities for each learning outcome***
4. Apply principles of safe medication prescribing and management.	Unit Examinations, In-class quizzes, Discussion Board assignments, Clinical Case assignments and Comprehensive Final
5. Communicate medication-related information effectively to patients and caregivers.	Unit Examinations, Discussion Board assignments, Clinical Case assignments and Comprehensive Final

Basis of Grade Determination*

The course will use a 1,000-point required assessment system. Each 1% of the course grade equals 10 weighted course points**.** Unit examinations assess unit-level knowledge and clinical application; quizzes provide lower-stake feedback; discussion boards and clinical cases reinforce comparison, therapeutic reasoning, safe medication management, and patient communication**.**

Required assessment	**Number**	**Total Weight**	**Total Weighted points**
Unit examinations (9% each)	6	54%	540
Comprehensive final examination (16%)	1	16%	160
In-class clinical pharmacology quizzes (1,667% each)	6	10%	100
Discussion board activities (2% each)	5	10%	100
Clinical case - based activities (2% each)	5	10%	100
TOTAL REQUIRED COURSE GRADE		100%	1,000

1. Unit Examinations — 54%
There will be six-unit examinations, each worth 9% of the course grade (90 weighted points). Each examination is planned to contain 50 questions. Questions may assess mechanisms, indications, contraindications, adverse effects, interactions, medication selection, dosing, monitoring, and patient education. Clinical vignettes may require integration of age, medical history, allergies, current medications, laboratory findings, and other patient-specific factors.

2. Comprehensive Final Examination — 16%
The comprehensive final is planned to contain approximately 50–60 questions and will assess

integration across Units 1–6. The following blueprint describes the percentage of the final examination devoted to each area. Passing requirement: students must earn at least 75% on the comprehensive final and at least 75% overall in the course. No remediation is allowed for the comprehensive final.

3. In-Class Clinical Pharmacology Quizzes — 10%
Six in-class quizzes are planned, one after each unit. Each quiz will contain 10 questions and have a 10-minute time limit. Quizzes assess foundational knowledge, clinical application, and medication safety/decision-making and provide feedback about areas requiring further study. Each quiz contributes approximately 1.667% of the course grade, for a total of 10% (100 weighted points). Attendance and missed-quiz rules will be applied consistently with program attendance policy and approved accommodations.

4. Discussion Board Activities — 10%
Five discussion board activities will be delivered through Brightspace. Each activity is worth 2% of the course grade (20 weighted points). Discussion activities are intended to support peer education, exchange of ideas, and clinical reasoning. Unless the activity instructions specify otherwise, students should complete their initial work independently before engaging with peers and must submit their own work. Each discussion prompt will identify the expected initial post, peer responses, deadline, rubric, and permitted resources. Late-work rules are subject to applicable program policy.

5. Clinical Case-Based Activities — 10%
Five case-based activities will be delivered through Brightspace. Each is worth 2% of the course grade (20 weighted points). Cases reinforce clinical decisions based on presentation and patient-specific factors, including dosing, administration, monitoring, and patient education. Each case will be graded using a rubric that addresses problem identification, therapy selection and rationale, contraindications and interactions, dosing/administration, monitoring, and patient counseling, as applicable to the case.

6. Optional Checkpoint Quizzes — Bonus
Students will have the opportunity to participate in taking eight optional checkpoint quizzes. Checkpoints are completed individually outside class through Brightspace and provide retrieval practice and immediate feedback. Each checkpoint allows two attempts.

Bonus rule*: Students will be awarded 3.75 bonus points for each checkpoint on which student earns at least 90% on either attempt, up to a maximum of 30 bonus points (3% of the 1,000-point course total). Bonus points are added after calculating the required assessment total, with the final course score capped at 1,000 points/100%.*

Grading scale; Percentage Points out of 1000; Letter Grade; Course Status

Percentage	Points out of 1,000	Letter grade	Course status
90%–100%	900–1,000	A	Excellent
80%–89%	800–899	B	Good
75%–79%	750–799	C	Pass
Below 75%	Below 750	F	Not passing

Final course grades will not be rounded. A student whose course grade is below 75% does not pass the course. Students should consult the PA Studies Program Guide for promotion and academic progression policies

Schedule:

Date	Class topic	Planned assignments / Reading Chapters/ Bonus Checkpoint Quizzes	Exam or In-Class Quiz
Thu, Oct 15	Unit 1: Introduction to Pharmacology	DB 1: Patient-Specific PTX, due Mon, Oct 19; Checkpoint 1 due, Wed, Oct 21 **Katzung's Reading Assignment**: Intro to Pharmacology, Unit 1, Chapters 1-5	—
Tue, Oct 20	No class	—	—
Thu, Oct 22	Unit 1: Introduction to Pharmacology	DB 2: Comparing Agents Within a Drug Class, due Mon, Oct 26; Checkpoint 2 due, Wed, Oct 28	—
Tue, Oct 27	Unit 2: Musculoskeletal	Case 1: Musculoskeletal PTX 1 due Nov 5 **Katzung's Reading Assignment:** Musculoskeletal, Unit 2, Chapters 27, 36, 42	**In-class virtual Quiz 1:** Thu, Oct 27 (8:35 am – 8:45 am via Lockdown Browser); **In-person Unit 1 Exam:** Fri, Oct 30, Rm# NAB2243 (8:30 am -9:30 am)
Thu, Oct 29	Unit 2: Musculoskeletal	DB 3: Medication Safety & Error Prevention, due Mon, Nov 2;	—

		Checkpoint 3 due, Wed, Nov 4	
Tue, Nov 3	Unit 3: GI/Nutrition	Case 2: GI & Nutrition PTX due Tue, Nov 10 **Katzung's Reading Assignment:** GI & Nutrition, Unit 3, Chapter 62 (Drugs Used in the Treatment of Gastrointestinal Diseases) and Chapter 65 (Dietary Supplements & Herbal Medications)	**In-class virtual quiz 2:** Tue, Nov 3 (8:35 am – 8:45 am via Lockdown Browser); **In-person-Unit 2 Exam**: Fri, Nov 6, Rm# NAB2243 (8:30 am -9:30 am)
Thu, Nov 5	Unit 3: GI/Nutrition	DB 4: Clinical Pharmacology Case, due, Mon, Nov 9; Checkpoint 4 due, Wed, Nov 11	—
Tue, Nov 10	Unit 4: Endocrine	Case 3: Endocrine PTX, due Thu, Nov 19 **Katzung's Reading Assignment:** Unit 4, Chapter 37-42 (Hypothalamic & Pituitary Hormes; Thyroid & Antithyroid Drugs; Adrenocorticosteroids & Adrenocortical Antagonist; Gonadal Hormones & Inhibitors; Pancreatic Hormones & Glucose-Lowering Drugs; Agents that Affect Bone Mineral Homeostasis)	**In-class virtual quiz 3:** Tues, Nov 10 (8:35 am – 8:45 am via Lockdown Browser); **In-person Unit 3 Exam:** Fri, Nov 13, Rm# NAB2243 (8:30 am -9:30 am)
Thu, Nov 12	Unit 4: Endocrine	DB 5: Patient Medication Counseling, due, Mon, Nov 16; Checkpoint 5 due, Wed, Nov 18	—
Tue, Nov 17	Unit 5: Renal	Case 4: Renal PTX, due Mon, Nov 23 **Katzung's Reading Assignment:** Renal, Unit 5, Chapter 15 (diuretic Agents) and Chapter 3 (Pharmacokinetics & Pharmacodynamics: Rational Dosing and the Time Course of Drug Action)	**In-class virtual quiz 4:** Tues Nov 17 (8:35 am – 8:45 am via Lockdown Browser); **In-person Unit 4 Exam:** Fri, Nov 20, Rm# NAB2243) (8:30 am -9:30 am)
Thu, Nov 19	Unit 5: Renal	Checkpoint 6 listed Tue, Nov 24	—
Tue, Nov 24	Unit 6: GU/Reproductive	Case 5: GU/Reproductive PTX, due Thu, Dec 3; Checkpoint 7 due, Wed, Dec 2 **Katzung's Reading Assignment:**	**In-class virtual quiz 5:** Tues Nov 24 (8:35 am – 8:45 am via Lockdown

		Genitourinary & Reproductive, Unit 6, Chapter 40 (The Gonadal Hormones and Inhibitors) and selected sections of Chapter 41 (Pancreatic Hormones & Antidiabetic Drugs), focused on the reproductive/metabolic overlap (e.g., PCOS and insulin resistance)	Browser)
Thu, Nov 26	No class — Thanksgiving break	—	—
Tue, Dec 1	Unit 6: GU/Reproductive	Checkpoint 8 due, Wed, Dec 9	*__In-class virtual Unit 5 exam:__ Tues, Dec 1: (8:35 am – 9:35 am via Lockdown browser) *Note: Exam date changed due to Thanksgiving break.*
Thu, Dec 3	Last Day of Class; open forum/discussion		**In-class virtual quiz 6:** Dec 3 (8:35 am – 8:45 am via Lockdown Browser); **In-person Unit 6 Exam:** Fri, Dec 4, Rm# NAB2243 (1:00 pm – 2:00 pm)
Tue, Dec 8	No class	—	—
Thu, Dec 10	**Comprehensive Final Examination**	—	**In-person Comprehensive Final Exam:** Fri, Dec 11 Rm# NAB2243 (1:00 pm – 2:00 pm)

Key: DB= Discussion Board; PTX = Pharmacotherapy; Checkpoint = Bonus Quiz

Any material covered in lecture or assigned as reading is important and is therefore testable.

** Note: The Schedule is subject to revision*

Readings, handouts, and Other Course Materials

Required Text

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- Katzung's Basic and Clinical Pharmacology, Sixteenth Edition. Todd W. Vanderah. 2025, McGraw-Hill Education, ISBN 1260463303 (eBook); https://accessmedicine.mhmedical.com/book.aspx?bookid=3382

Recommended Text

- Pharmacotherapy Principles and Practice. Chisholm-Burns, Schwinghammer, Malone, Kolesar, Lee. 2022, McGraw-Hill Education. Print ISBN 9781260460278; eText ISBN 9781260460285. https://suny-ups.primo.exlibrisgroup.com/permalink/01SUNY_UPS/51sa6u/alma991128037304861

Relevant Policies, Practice, and Statements

- **Course Specific Policies**
  None

- ***Program Specific Policies***
  Please refer to the Department of PA Studies Student Program Guide for additional information on attendance, policy violations, grading and academic progression.

- ***Academic Honesty/Code of Conduct***
  https://www.upstate.edu/student-handbook/conduct.php

- ***Class Cancellation***
  In the event of class cancellation due to unforeseen circumstances such as instructor illness, emergencies, or inclement weather, students will be notified as promptly as possible through official university communication channels, including communications from the course director who may arrange for remote same-day instruction or other alternatives.

  Students are responsible for regularly checking their university email for updates and should communicate with the course director if they have any questions or concerns.

- ***Religious Holidays***
  Students must alert the course instructor the **first week of the course** of any religious observances or holidays that will affect class attendance or assignment dates. Class and Religious Beliefs Policy

- ***ADA and Access Statement***
  Upstate is committed to creating a learning environment that meets the needs of its diverse

student body. Students who anticipate/experience barriers to learning in this course, such as health events (ex. Illness, injury), or significant life events (e.g., a loss, a caregiving emergency) should notify the Course Director ASAP.

If you have or think you may have a disability, please schedule a meeting with our Disability Support Services Office to begin a conversation or request an official accommodation as soon as possible. If you have already been approved for accommodations through Disability Support Services, please meet with your instructor or course coordinator to review your accommodations letter and develop an implementation plan.

- ***Suggested Learning Environment with Diversity and Inclusion Statement***
  We strive for an environment that supports inquiry, curiosity, trust, meaningful discussion, and learning. Our intent is to respect divergent perspectives and honor your evolving professional identity as future health care professionals, including contributing factors such as race, gender, class, sexuality, religion, ability, and others. We ask that learners and faculty collaborate in creating this environment.

  Please contact us for concerns that you wish to discuss, no matter how small they may seem. We are always willing to listen and support learners.

- ***Student Complaint Policy***
  Student concerns about a course, course policy, grading should first be brought to the course director. Each college has a process for reviewing concerns about grades. Any student who would like to register a complaint regarding Upstate Medical University or any of its schools or colleges may file a formal complaint with the Dean of Student Affairs within 90 days of the occurrence or event giving rise to the grievance. Students may also file a complaint anonymously via the Student Complaint Website.

- ***Additional Policies found in the Student Handbook***
  Policies & Expectations | Student Handbook | SUNY Upstate

- ***Statement on the use of Artificial Intelligence (AI)***
  The University holds no specific position endorsing or restricting the acceptable use of AI tools in coursework, within the prevailing principles of academic freedom, honesty and integrity specified in the Code of Student Conduct.

  Each assignment should also include instructions on the use of AI/machine learning for that assignment.

  Additional information on the Use of AI in course work can be found at the University of Delaware Center for Teaching and Assessment of Learning at
  https://sites.udel.edu/ctal/advanced-automated-tools/#syllabus-language .

Other Student Resources

Student basic needs and available support services:

Upstate recognizes that students' ability to do well in courses is affected by their academic skills and by access to basic needs (food, housing, transportation, healthcare, mental health support, access to technology, and emergency financial assistance). We commit to support each student in achieving their educational goals by providing resources to help meet basic needs.

Asking for help can be difficult. Seeking support reflects strength and a proactive approach to managing your well-being. If you are experiencing difficulties, please reach out. All discussions will remain confidential, and our primary goal is to help you succeed.

Offices and functions to support students and student achievement are coordinated through the Office of Student Affairs. These include student support services, disability services, student counseling, student health services, services to help with food insecurity, a wellbeing and mental health resource center, and many others.